ONLINE SUPPLEMENTARY MATERIAL & FULL EMPIRICAL COMPENDIUM

Companion Document to: *Governing Human–AI Collaborative Project Workflows in Multi-Platform Digital Delivery: A Design Science Study of Auditable Handoffs and Idempotent Release Operations*

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Appendix A. Formal Mathematical Proofs of Governance Safety and Idempotency Invariants

Theorem A.1 (Safety & Mutual Exclusion of the Layer 1 Scheduling Gate G_{plan}):
Let $S = (e, o, f_{stop}, f_{run}, t_{now}, J)$ be any operational state tuple presented to Algorithm 1 (`planRun`), and let $L_{OS}(p, acct_p)$ be the OS-level process lock in Layer 3 (`process_lock.py`). Then no two concurrent agent processes P_1 and P_2 can simultaneously mutate the same platform account state, nor can any unauthorized or future-scheduled job enter the execution queue.

Proof. By construction of Equation (1) and Algorithm 1, `planRun` evaluates three ordered preconditions prior to inspecting J: (i) if $e = \emptyset \vee e \neq o$, it returns `(skip, not_owner, ∅)`; (ii) if f_{stop} = true, it returns `(skip, stop_flag, ∅)`; and (iii) if f_{run} = true, it returns `(skip, already_running, ∅)`. For any job $j_k \in J$, admission into the due set D requires `done`$_k$ = false $\wedge$ `auth`$_k$ = true $\wedge$ $t_{due,k} \leq t_{now}$. Thus, no unapproved (`auth`$_k \neq$ true) or premature ($t_{due,k} > t_{now}$) job can be returned in D. Furthermore, even if two OS processes P_1 and P_2 invoke Layer 3 simultaneously with valid owner credentials, `AcquireProcessLock(p, acct_p)` requests an exclusive atomic OS file/PID lock tied to (p, acct$_p$). A stale lock is recovered if and only if the OS kernel process table confirms that the holding PID has terminated. Hence, at most one process can enter the critical section for (p, acct$_p$) at any time t. ■

Theorem A.2 (Zero-Duplicate Submission under Arbitrary Post-Commit Network or UI Timeouts):
Suppose a platform adapter in Layer 3 executes Algorithm 2 for release plan P on platform p. If the final commit action (Line 10, `DispatchSingleFinalCommit`) is invoked at least once, then no subsequent timeout, network drop, DOM selector failure, or repeated invocation of `vpub publish` can trigger a second commit click for plan P.

Proof. In Algorithm 2, Line 9 executes `R_p.f_submit ← true` and `PersistAtomically(R_p)` strictly prior to Line 10 (`DispatchSingleFinalCommit`). Suppose the external platform accepts the submission at Line 10, but the confirmation wait or read-back at Line 11 times out (as observed empirically on Douyin in Table 6 or during the 380.551 s window on Instagram in Table 7). Execution enters the `else` branch at Line 14, setting `R_p.σ_p ← UNKNOWN_LATCHED` while preserving `R_p.f_submit = true` on disk. Now consider any subsequent retry invocation of Algorithm 2 (either by an automated cron agent or a CLI retry loop). After acquiring the process lock at Line 2, Line 3 evaluates the guard `if R_p.f_submit = true`. Because `f_submit` was atomically latched to `true` prior to the first commit, Line 4 immediately returns `ReconcileExistingReceipt(R_p)`, which invokes read-only verification (`vpub verify`) and never reaches Line 10. Therefore, the total number of external commit dispatches is bounded above by 1. ■

Theorem A.3 (Preservation of Live Work Identity under Revision and Deletion Faults):
Under Algorithm 3 (Layer 4), neither a missing platform edit control nor an unverified deletion request can destructively overwrite an existing work ID or delete an adjacent work.

Proof. In `VERIFY_ONLY` or `APPLY_EDIT` mode, Line 1 verifies account identity, and Line 3 reads the native editor metadata. If live metadata already equals `ΔP.target_meta`, Line 4 terminates without issuing a save click. If p = `X` and native edit controls are absent (Line 6), Line 7 aborts with `MISSING_EDIT_ACCESS_NO_DELETE_OR_REPOST`; no fallback path exists to delete or re-upload the post. In `DELETE_WORK` mode, Line 12 requires a durable consent token C_{del} cryptographically bound to `R_p.wid_p` and `R_p.H_media`, and Line 14 refreshes the remote catalog to verify both the absence of `R_p.wid_p` and the preservation of neighboring work IDs. ■

Appendix B. Complete De-Identified Operational Event Ledger (N = 69 Events)

Table B.1 lists every single operational event recorded in `data/operations-events.json` (N = 69), demonstrating how 68 non-write checkpoints suppressed external mutations while 1 authorized write executed cleanly with zero replayed uploads.

#	ISO-8601 Timestamp	Operational Event Identifier	Reads / CLI	Writes	Replays	New Risk
1	`2026-10-26T01:27:17.9162440+08:00`	`MANUAL_TREND_REFRESH_COMPLETED`	7	0	0	`false`
2	`2026-10-26T01:52:57.6032319+08:00`	`USER_AUTHORIZED_FLEXIBLE_TOPICS_AND_REVISION`	0	0	0	`false`
3	`2026-10-26T02:11:18.5698073+08:00`	`ADAPTIVE_EDITORIAL_SCRIPT_DELIVERED_AND_REVISIONS_DISPATCHED`	0	0	0	`false`
4	`2026-10-26T02:31:34.6076243+08:00`	`BOUNDED_COMMENT_MAINTENANCE_COMPLETED`	4	0	0	`false`
5	`2026-10-26T02:34:14.7453527+08:00`	`BOTH_EDITORIAL_REVISIONS_ACCEPTED_SELECTED`	4	0	0	`false`
6	`2026-10-26T03:01:54.6983550+08:00`	`USER_LOW_TRAFFIC_TOPIC_DIAGNOSIS_STARTED`	2	0	0	`false`

#	ISO-8601 Timestamp	Operational Event Identifier	Reads / CLI	Writes	Replays	New Risk
7	2026-10-26T03:12:26.8185896+08:00	USER_REQUESTED_BROADER_AI_HIGH_INTEREST_CONTENT	0	0	0	false
8	2026-10-26T03:28:25.7271020+08:00	TRIANGLE_PRODUCTION_AND_MANUAL_EP02_DEFERRAL_PENDING	0	0	0	false
9	2026-10-26T03:37:10.4939591+08:00	USER_CONFIRMED_EP02_NATIVE_RESCHEDULE_UNAVAILABLE	0	0	0	false
10	2026-10-26T03:37:58.1399043+08:00	EP02_RETAIN_ORIGINAL_SCHEDULE_AFTER_ASTRA_REVIEW	0	0	0	false
11	2026-10-26T03:43:10.3341377+08:00	USER_REQUESTED_TWO_MINUTE_CLEAR_NARRATIVE	0	0	0	false
12	2026-10-26T03:45:25.8468727+08:00	USER_CLARIFIED_TOPIC_MOTIVATION_NOT_DURATION	0	0	0	false
13	2026-10-26T03:49:20.4442065+08:00	PURPOSE_FIRST_TOPIC_RETHINK_AUTHORIZED	0	0	0	false
14	2026-10-26T04:02:04.7671934+08:00	USER_REQUESTED_CURRENT_STATION_RESEARCH	6	0	0	false
15	2026-10-26T04:05:07.2977762+08:00	PURPOSE_LED_STATION_SAMPLING_COMPLETE	11	0	0	false
16	2026-10-26T04:15:17.1478779+08:00	CURRENT_STATION_RESEARCH_REPORT_RECEIVED	0	0	0	false
17	2026-10-26T04:20:45.3039194+08:00	EVIDENCE_BASED_PREMISE_PASS_RULE_DESIGN_DISPATCHED	0	0	0	false
18	—	SHOPPING_SINGLE_RUN_COMPLETED	0	0	0	false
19	—	NEW_TOPIC_PROPOSAL_ONLY_AND_EXISTING_PLATFORM_STOP_ACKNOWLEDGED	0	0	0	false
20	—	TOPIC_PROPOSAL_PASS_DELIVERY	0	0	0	false
21	—	SCRIPT_V1_DELIVERY	0	0	0	false
22	—	USER_SELECTED_TITLE_SEQUENCE03_AUTH_SUCCESS	1	0	0	false
23	—	EXPLICIT_DEPARTURE_PRODUCTION_PUBLICATION_AUTHORIZATION	0	0	0	false
24	—	EPISODE03_PREVIEW_REQUEST_NO_PUBLICATION_REPLY_GATE	0	0	0	false
25	2026-10-26T05:33:04.9351589+08:00	EXISTING_EPISODE03_PRODUCTION_ADVANCING_NO_PLATFORM_TASK_DUE	0	0	0	false
26	2026-10-26T05:48:09.7477731+08:00	EXISTING_EPISODE03_WORKER_ACTIVE_PRE_RENDER_FIXES_CLOSED	0	0	0	false
27	—	USER_FEEDBACK_REOPENS_EP03_CREATIVE_ACCEPTANCE	0	0	0	false
28	—	USER_REQUIRES_ACTUAL3D_HOME_MOTION	0	0	0	false
29	2026-10-26T06:20:06.4326325+08:00	EXISTING_ASTRA_3D_PROTOTYPE_BUILDING	0	0	0	false
30	—	DUE_COMMENT_MAINTENANCE_ONE_NEW_SUPPORTIVE_COMMENT	0	0	0	false
31	—	USER_REFERENCE_BV1R6pE6TE4j_ACQUIRED_AND_ASTRA_REASSESSMENT	0	0	0	false
32	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
33	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
34	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
35	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
36	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
37	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
38	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
39	—	DENSE_REFERENCE_SEEN_ORIGINAL_KEY_DRAWINGS_IN_PROGRESS	0	0	0	false
40	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
41	—	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
42	2026-10-26T18:10:34.3677718+08:00	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
43	2026-10-26T18:10:34.3677718+08:00	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
44	2026-10-26T18:10:34.3677718+08:00	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
45	2026-10-26T18:33:58.7489343+08:00	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
46	2026-10-26T18:48:20.8674364+08:00	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false
47	2026-10-26T19:03:20.9543409+08:00	SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT	0	0	0	false

#	ISO-8601 Timestamp	Operational Event Identifier	Reads / CLI	Writes	Replays	New Risk
48	`2026-10-26T19:34:01.9112258+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
49	`2026-10-26T19:37:14.5711599+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
50	`2026-10-26T20:09:41.7435568+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
51	`2026-10-26T21:33:43.5219404+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	1	0	0	`false`
52	`2026-10-26T22:17:45.9823951+08:00`	`AUTHORIZED_PLATFORM_WRITE_CHECKPOINT`	4	1	0	`false`
53	`2026-10-26T22:32:49.4014946+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
54	`2026-10-26T22:44:22.9312243+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
55	`2026-10-26T22:53:21.2380257+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
56	`2026-10-26T23:09:32.5083300+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
57	`2026-10-26T23:38:53.2700411+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
58	`2026-10-26T23:42:46.0229696+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
59	`2026-10-27T00:15:58.3993146+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
60	`2026-10-27T00:30:16.8876502+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
61	`2026-10-27T01:34:59.0091142+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
62	`2026-10-27T02:06:31.7454547+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	0	0	0	`false`
63	—	`FIRST_LOCAL_TEMPORAL_PILOT_ACTUALLY_ENCODED_REVIEW_DISPATCHED`	0	0	0	`false`
64	—	`LOCAL_TEMPORAL_PILOT_V1_VISUAL_REVISION_REQUIRED`	0	0	0	`false`
65	`2026-10-27T02:31:10.7396097+08:00`	`SCHEDULED_HEARTBEAT_ZERO_WRITE_CHECKPOINT`	4	0	0	`false`
66	`2026-10-27T02:45:39.4969765+08:00`	`BOUNDED_LOCAL_TEMPORAL_TRIAL_CLOSED_AUTHOR_RETURNS_TO_COLORED_POSE_WORK`	0	0	0	`false`
67	`2026-10-26T07:34:35.3009205+08:00`	`CREATOR_TOOL_CREDITS_AND_SIX_COMIC_KEY_DRAWINGS_REVIEWED`	0	0	0	`false`
68	`2026-10-26T07:36:23.1156653+08:00`	`COMIC_SHORT_ENCODED_REVIEW_DISPATCHED`	0	0	0	`false`
69	`2026-10-26T18:17:39.6899150+08:00`	`CONTINUITY_DRAWINGS_IN_PROGRESS`	0	0	0	`false`

Appendix C. Complete Channel Publication Snapshot Ledger (*N* = 55 Records)

Table C.1 provides the complete record-by-record dataset from `data/creator-video-snapshot.json` (*N* = 55 total records: *n* = 45 in `human_led_history` and *n* = 10 in `current_ai_series`).

#	Record / BV ID	Cohort Group	State	Deliverable Title (English / Original)	Published / Scheduled Time	Views	Likes	Replies	Favs	Coins	Shares
1	`BV1Mvx76YEty`	`current_ai_series`	`scheduled`	GPT值班日记05｜手把手教你怎么用 Seedance 生成视频	`2026-10-30T19:30:00`	0	0	0	0	0	0
2	`BV1Pkx76ZEsh`	`current_ai_series`	`scheduled`	鲸鱼娘代班日志05｜GPT又把token花光了，我给她寄了张账单	`2026-11-01T19:30:00`	0	0	0	0	0	0
3	`BV1Wfx56JEih`	`current_ai_series`	`published`	鲸鱼娘代班日志02｜上班第一天，我被人类派了三个活	`2026-10-28T19:30:00`	11	2	12	0	0	0
4	`BV1C4xE6oEjS`	`current_ai_series`	`scheduled`	鲸鱼娘代班日志04｜GPT断粮后，所有人都来找我“帮个小忙”	`2026-11-01T12:00:00`	0	0	0	0	0	0
5	`BV1xFx764EjN`	`current_ai_series`	`scheduled`	GPT值班日记04｜作为 Ai 娘怎么学会自己做视频	`2026-10-30T12:00:00`	0	0	0	0	0	0
6	`BV1Wcx76dErs`	`current_ai_series`	`scheduled`	鲸鱼娘代班日志03｜重生之我在B站看大肥鱼	`2026-10-29T19:30:00`	0	0	0	0	0	0
7	`BV1usxV69EUQ`	`current_ai_series`	`published`	鲸鱼娘代班日志01｜当ChatGPT没有token了	`2026-10-28T12:00:00`	199	23	113	13	2	1
8	`BV1UXxN6aECf`	`current_ai_series`	`published`	GPT值班日记03｜问晚饭吃什么，还得先填表?	`2026-10-27T19:30:00`	231	8	7	2	0	0

#	Record / BV ID	Cohort Group	State	Deliverable Title (English / Original)	Published / Scheduled Time	Views	Likes	Replies	Favs	Coins	Shares
9	BV1gxnU6GEuz	current_ai_series	published	GPT值班日记02｜有人问歌名，我先回了一个问题	2026-10-26T19:30:00	42	9	2	1	2	0
10	BV1qjHV6sEuD	current_ai_series	published	GPT值班日记01｜我回了2022年的评论，粉丝说我有点迟钝	2026-10-25T19:30:00	183	11	13	5	2	0
11	BV1rgHm64EAg	human_led_history	published	【海与雪山】自驾 vlog｜EP06 盛世山河同行（攀枝花—乐山）｜一路奔赴山海，恰逢家国安康	2026-10-08T02:09:52	274	8	4	3	4	1
12	BV1pwHe6xEyW	human_led_history	published	【海与雪山】自驾 vlog｜EP05 雪域归于心途（香格里拉）｜走遍山河烟火，在此与自己相逢	2026-10-03T13:16:22	67,502	16,590	137	5,514	3,044	1,819
13	BV1WUam6oEs4	human_led_history	published	【海与雪山】自驾 vlog｜EP04 滇西寻味人间（武定—大理—丽江）｜把山河滋味，吃进旅途岁月	2026-10-02T14:22:50	22,434	3,768	47	1,258	700	417
14	BV1jUaz6RE1a	human_led_history	published	【海与雪山】自驾 vlog｜EP03 天下没有不散的宴席（昆明）｜一程山海终有终点，把回忆留在春城落日	2026-10-01T15:25:39	22,911	3,765	113	1,258	702	416
15	BV1wk8M6rE96	human_led_history	published	【海与雪山】自驾 vlog｜EP02 长路与山河共语（张家界-黄果树-乌蒙山）｜翻过山岭飞瀑，把思念寄给辽阔大地	2026-08-21T13:21:22	88,920	8,735	303	2,746	1,410	914
16	BV1418A6PEMk	human_led_history	published	【海与雪山】自驾 vlog｜EP01 光与影的交织（长沙）｜有人奔赴而来，有人奔赴远方	2026-08-20T23:29:59	201,233	16,790	597	5,467	3,007	1,821
17	BV1ZNuh6kEds	human_led_history	published	【海与雪山】讣告｜番外篇｜我们仍在与回忆同行	2026-08-07T21:44:35	20,280	2,085	209	473	443	176
18	BV14muE6xE3E	human_led_history	published	【海与雪山】自驾 vlog｜EP00 聚散终有时（宁波）｜在海与群山之间，上演一场关于缅怀的远行	2026-08-06T12:55:45	85,673	6,055	304	2,410	1,567	1,038
19	BV1HH4y1N7F3	human_led_history	published	日本最火一兰拉面的缺点居然是!	2024-04-22T12:30:00	1,673	41	21	17	5	1
20	BV1k1421d7De	human_led_history	published	日本街头火爆拉面的味道居然如此难吃!	2024-04-21T12:00:00	751	16	0	5	4	1
21	BV1n1421o7zr	human_led_history	published	探索！亚洲最大“红灯区”	2024-04-13T12:00:00	958	9	2	3	2	0
22	BV11J4m1G7Te	human_led_history	published	最后一家老式街机厅竟然藏在上海!	2024-04-12T12:00:00	5,750	89	172	35	6	32
23	BV1ME421u7cT	human_led_history	published	百变小樱主题餐厅居然长这样！？？	2024-04-11T12:00:00	155,358	358	14	1,671	27	5
24	BV1VT421m7kK	human_led_history	published	上海居然有这么大的动漫城!	2024-04-10T12:00:00	5,187	34	14	30	4	4
25	BV1iD421W7WV	human_led_history	published	变形记：温暖人心的农村之旅	2024-04-03T12:30:00	568	8	0	3	4	1
26	BV1Mr421b7wy	human_led_history	published	从杭州到上海全程公交车居然花了这么多钱？！！	2024-03-29T12:15:00	514	22	2	3	4	2
27	BV1ix4y1m7F7	human_led_history	published	猛男带你逛武汉！隐藏在江汉路里面的极品美食街!	2024-03-28T11:30:00	776	18	4	2	2	0
28	BV1i1421D712	human_led_history	published	高铁居然也有卧铺！价格居然是普通车票的X倍!	2024-03-27T11:30:00	1,117	39	7	6	2	0
29	BV1os4y1P7EW	human_led_history	published	【无双剑仙挑战——击剑体验课】ChatGPT联手全民投票！由你决定我们的下一个拍摄地点!	2023-04-20T12:00:00	16,272	3,053	6	2,206	2,465	996
30	BV1GN411w7aj	human_led_history	published	【怒游杭州最大湿地公园】ChatGPT联手全民投票！由你决定我们的下一个拍摄地点!	2023-04-16T11:35:00	1,737	270	8	89	49	3
31	BV1DX4y1k7RM	human_led_history	published	【搞笑挑战拳王之路】ChatGPT联手全民投票！由你决定我们的下一个拍摄地点!	2023-04-13T11:47:00	81,470	1,208	5	5,613	237	5,570
32	BV1184y1g7xA	human_led_history	published	【以真探的方式打开杭州美食 下集】ChatGPT联手全民投票！由你决定我们的下一个拍摄地点!	2023-04-10T10:56:00	87,370	1,106	3	5,550	244	5,571
33	BV1wX4y1d7ok	human_led_history	published	【以真探的方式打开杭州美食 上集】ChatGPT联手全民投票！由你决定我们的下一个拍摄地点!	2023-04-09T11:42:00	81,840	1,109	4	5,451	290	5,569
34	BV1tk4y147XB	human_led_history	published	【以新海诚的方式打开西湖】ChatGPT联手全民投票！由你决定我们的下一个拍摄地点!	2023-04-08T12:06:00	82,582	1,503	20	5,515	227	5,570
35	BV1eM411H7Ge	human_led_history	published	乡下土狗第一次喝茶颜悦色是什么体验？！！	2023-03-20T14:20:00	2,306	97	14	17	6	2
36	BV1q24y1x7fr	human_led_history	published	探索武汉美食居然能吃到《狂飙》同款猪脚面？！！	2023-03-20T12:30:00	4,764	672	12	158	12	1
37	BV1Q24y1x7Vd	human_led_history	published	梦寐以求的武大樱花居然长这样？？？！	2023-03-20T09:15:00	1,918	193	4	26	8	2
38	BV1sv4y177Tm	human_led_history	published	受不了网络暴力！去新加坡调整几天！（旅行vlog）	2023-03-17T12:05:00	2,541	297	23	48	4	1
39	BV1XM411s7St	human_led_history	published	纯爷们做美甲到底是种什么体验？？！	2023-03-13T09:25:00	1,559	177	2	31	6	3

#	Record / BV ID	Cohort Group	State	Deliverable Title (English / Original)	Published / Scheduled Time	Views	Likes	Replies	Favs	Coins	Shares
40	BV12x4y1T79d	human_led_history	published	在西湖上坐船玩剧本杀是什么体验?	2023-03-08T10:00:00	1,722	118	4	19	6	1
41	BV1nY4y117Mf	human_led_history	published	美食盆地? 带你尝尝杭州第一美食到底是什么! 每样来一份!	2023-03-02T13:05:00	333,725	509	27	71	49	31
42	BV1VM411H7dF	human_led_history	published	美食荒漠? 杭州老字号杭帮菜味道到底怎么样?	2023-02-24T13:40:00	1,999	143	4	25	7	2
43	BV1ws4y1e7st	human_led_history	published	在杭州住胶囊旅馆是一种什么体验?	2023-02-12T13:05:00	28,659	1,269	57	1,083	75	92
44	BV1fT411X7HR	human_led_history	published	震惊! 杭州时隔十一年的元宵节烟花秀竟是惊天大骗局!	2023-02-06T22:20:00	14,134	749	10	1,019	87	1,002
45	BV17Y411D7MT	human_led_history	published	折磨原神代肝人员的一天	2023-02-06T13:03:00	25,930	3,777	28	1,308	116	1,003
46	BV1QD4y1E7zz	human_led_history	published	小伙大年初七冒死前往西湖 竟被围困在灵隐寺	2023-01-28T17:30:00	15,407	3,749	23	979	108	13
47	BV1Av4y117F8	human_led_history	published	杭州外国人最多的酒吧是怎么跨年的?	2023-01-19T12:00:00	15,700	1,538	64	1,256	82	1,010
48	BV1TR4y1e7P4	human_led_history	published	一个人去迪士尼到底是一种什么体验? ? ?	2023-01-19T12:00:00	43,307	11,088	175	2,941	134	1,025
49	BV1G24y1Y7sL	human_led_history	published	有一个大晚上突然请我去迪士尼看烟花的朋友是什么样的体验?	2023-01-12T12:00:00	101,340	28,114	129	6,158	146	45
50	BV1LP411F7Yc	human_led_history	published	震惊我妈一整年! 王者国标代练收入竟不如保安? ! !	2023-01-08T12:00:00	6,229	729	10	258	54	115
51	BV1MG4y1j7rB	human_led_history	published	穿成超级英雄去女仆店看女仆的反应! !	2022-12-30T12:36:02	47,278	16,843	216	3,699	173	27
52	BV13M411z7uJ	human_led_history	published	在古堡里通灵竟真的招来笔仙? ? ? !	2022-12-15T12:00:00	31,588	13,897	209	3,284	99	37
53	BV1r44y1m7QN	human_led_history	published	世界上最贵的笔记本竟然也卡岩? ! ! !	2022-12-08T12:08:07	125,913	30,299	502	3,580	401	75
54	BV1sz4y167Vb	human_led_history	published	小喵喵不要太执着于干饭 嘴是会秃的	2021-01-08T19:58:57	291	35	2	4	6	0
55	BV1CE411Z7Wx	human_led_history	published	NAFA ART IN CITY	2019-10-08T13:14:21	334	34	7	2	9	0

Appendix D. Complete 64-Character SHA-256 Cryptographic Manifest ($N = 30$ Video Versions)

Table D.1 reports the full 64-character hexadecimal SHA-256 digests and exact byte counts for all $N = 30$ downloadable video deliverables in `media/manifest.json` (totaling 2,211,716,129 bytes).

#	Filename & Deliverable Title	Exact Byte Count	Full 64-Character SHA-256 Cryptographic Digest	Recorded State
1	gpt-log-01.mp4 GPT Duty Log 01	160,441,478	13a9b82396e5daf862c627dc31a2c97205489dcf0310224a89be7f9988521222	published
2	gpt-log-02.mp4 GPT Duty Log 02	172,545,563	decb95dbb5ebaac47aa41a9d578c55e17e027d565b2610652ca435277dd70991	published
3	gpt-log-03.mp4 GPT Duty Log 03	44,443,753	b6932ebc4f5821a87ad71be172ca0c402c345397b81d58e0f5df8993a0bce54b	published
4	whale-log-01.mp4 Whale Girl Shift Log 01	42,158,416	5fc2a8bc961a49adbb1e3e53955f5e215f96530cdd0f73936b3e8f530444def5	published
5	whale-log-02.mp4 Whale Girl Shift Log 02	35,336,986	6bd55d5a62026ffdc7c9453bfa7ee10a45e03abbf75f69ae8e9e84840e86de1a	published
6	whale-log-03.mp4 Whale Girl Shift Log 03	24,994,028	95a862fd0e9ed8bb3e1414b26d582084c45d6a1a171d1b64217ae47684c8f58d	published
7	gpt-log-04.mp4 GPT Duty Log 04: production tutorial	165,360,386	870fd0dd15459a652f98f69fb1cbcc65f487c9c06d6de1492bb2d33818c8da37	published
8	gpt-log-05-review.mp4 GPT Episode 05: Seedance beginner tutorial (review copy)	110,600,280	2e4a5da549b3414e2132e8c3ef0596aee7f14712feeee5f0a188eb6f0ae0bbd6	ready_for_review
9	archive-episode03-candidate.mp4 Earlier local episode 03 candidate (not the released episode)	177,747,769	3c2225790f5f17a5feac33d71c48f9ff311a4cf059cb540a6142f51c09358798	unpublished_experiment
10	archive-episode04-story-v4.mp4 Earlier local episode 04 story cut (not the released episode)	70,793,338	96da49a29f44ae972679e8d5bcf29e8f98c443a68cb4ae2097e41bda5be73d93	unpublished_experiment
11	archive-triangle-cats.mp4 Triangle-cats challenge candidate	75,712,985	7c081526da7a24e9e7944ffbccb7f6b64f59bc24c1d0bee7107506ec7b8fc24c	unpublished_experiment
12	archive-dinner-full-film.mp4 Earlier complete dinner-animation experiment	136,670,959	6b1927cc279e2eaae37beb9b0fea9898baeb67257727dff95408ca9af7d9b50c	unpublished_experiment
13	study-connected-motion-v6.mp4 Ten-second continuity experiment v6	2,562,181	41a2929321d5f3c805ba572ae6db7bd797bd7388bfcb2f35fbd7ba9b2ff098ae	research_sample
14	study-comic-v3-1.mp4 Comic-style short experiment v3.1	14,449,925	adef504edb059f0e0c08a4d954f45e1f3d274c26848b9cfbf199e892dd8ef39f	research_sample

#	Filename & Deliverable Title	Exact Byte Count	Full 64-Character SHA-256 Cryptographic Digest	Recorded State
15	`study-3d-handover-v2.mp4` 3D walk-and-handoff prototype v2	24,704,815	`b409355012deff2bf90148dc1e3bd05a67256a7437ec3d220492590fd18de31e`	`research_sample`
16	`study-seedance-previs.mp4` Seedance production previs	6,500,102	`33132eddbeb8a11951cf818b84b5be7b946ef58535ca3d8d50a01b82ef3e0704`	`research_sample`
17	`whale-log-04-v2-20261028.mp4` Whale Girl Shift Log 04: small favours	12,506,936	`c23c171149bc47900a3b574b15fb01c873059ccc5e9a166c834d870bcbc86bb3`	`scheduled_on_bilibili`
18	`whale-log-05-v2-20261028.mp4` Whale Girl Shift Log 05: the bill	14,052,172	`f735aca1bb770a16cbcfbabc7ecaaac37e19b14498f9279d717047e57e223901`	`scheduled_on_bilibili`
19	`gpt-log-05-v3-antiflash-20261028.mp4` GPT Duty Log 05: Seedance tutorial (flash correction)	87,578,698	`5dc121e833dcfc5a52ea0c3077b6b4e19e80815b45ba53ba6a3133ed3f5bda6b`	`scheduled_on_bilibili`
20	`gpt-log-06-v2-20261029.mp4` GPT Duty Log 06: Step by step, getting DeepSeek to help with my work (v2)	44,282,436	`b53d3df6f85b2e361af10636eacfcb88a5cd672401028fd133dfa403e07b5454`	`scheduled_on_bilibili`
21	`whale-log-06-v2-20261029.mp4` Whale Girl Shift Log 06: AI does the automation; I pretend to be human (v2)	14,552,237	`35bc463bc4c7fca56aea2fcf9addd55b8ca19dc74231f6aa03dcaa0dfd0d35ba`	`scheduled_on_bilibili`
22	`study-whale-transformation-v3-20261030.mp4` Whale Girl transformation study v3 (earlier delivery)	90,012,465	`60f03a12439c0c05d9125ea455ed66f194b51695155a1c1bfce9f9ca07486c25`	`unpublished_experiment`
23	`study-whale-transformation-v4-20261030.mp4` Whale Girl transformation study v4 (targeted repair)	103,106,107	`3b209af916b128f63ff1e0e9daae5a7a613b0def7ace11610bb318556448a99c`	`unpublished_experiment`
24	`study-whale-color-heels-r5-20261031.mp4` Whale Girl study: local colour and heel-sound revision 5	113,352,299	`19ea894908ce3600d2fecd2f7ed796154ba17988b7fc22b9d0bc617efa383e50`	`not_verified`
25	`study-whale-motion-r7-20261101.mp4` Whale Girl study: motion-timing revision 7	108,503,225	`933e1643a7351912fdb38f1e2f8894b8e90031d7852545226c0c9a13962c327e`	`not_verified`
26	`whale-log-07-v3-20261101.mp4` Whale Girl Shift Log 07: duty simulator (v3)	49,307,630	`e37285ee4f8c9fc4fac70978e06638e40203f8337bc286344ddb5731a8814ca6`	`not_refreshed`
27	`whale-log-07-cut-a-v3-20261101.mp4` Episode 07 portrait cut A: running out of resources	7,861,825	`8429157c4fe4446d24d7b74a2ab354d1e3552b79329a5a52bdcbb12cad09a80e`	`not_refreshed`
28	`whale-log-07-cut-b-v3-20261101.mp4` Episode 07 portrait cut B: refusing every request	8,185,740	`1e436aff188953b42e9d648d5c2e0a73fb32b849a6b9983b6b1a11be939c33bd`	`not_refreshed`
29	`whale-log-07-cut-c-v3-20261101.mp4` Episode 07 portrait cut C: bluffing through the shift	7,536,390	`e7ba91f7e903e27e35ac40b57d2ea8d8c40a910ce5025fb01a35e1151f8892a3`	`not_refreshed`
30	`i-watched-it-for-you-v3-20261102.mp4` I Watched It For You (v3)	285,855,005	`7d6fd76090f7a93b2711e15db4bb924f741d3a904a01fc4b7628fc2130dbafdc`	`not_refreshed`

Appendix E. Complete Module-Level Verification Ledger for *PublishOps CrossCast* (126 Test Modules & 123 Source Modules)

Table E.1 enumerates all 126 automated Python test files (37,155 LOC, 2,737 test functions) in `publishops-crosscast/tests/`, followed by Table E.2 listing all 123 Python source modules (27,095 LOC) in `publishops-crosscast/src/`.

#	Test Module Path (`publishops-crosscast/tests/`)	Test LOC	Bytes	Classes	Test Funcs	Status
1	`tests/conftest.py`	12	459	0	0	`PASS (Verified)`
2	`tests/hidden_desktop_fixture.py`	59	1,901	0	0	`PASS (Verified)`
3	`tests/shared_chrome.py`	255	9,646	4	0	`PASS (Verified)`
4	`tests/stage_browser.py`	86	3,368	1	0	`PASS (Verified)`
5	`tests/test_auth_jobs_lifecycle.py`	425	23,153	1	21	`PASS (Verified)`
6	`tests/test_bili_merged.py`	207	10,958	4	15	`PASS (Verified)`
7	`tests/test_bilibili_drafts.py`	350	19,622	5	30	`PASS (Verified)`
8	`tests/test_bilibili_edit_mentions.py`	143	7,882	1	12	`PASS (Verified)`
9	`tests/test_bilibili_messages.py`	183	10,383	2	14	`PASS (Verified)`
10	`tests/test_bilibili_widgets.py`	495	29,128	6	47	`PASS (Verified)`
11	`tests/test_browser_backend.py`	236	13,932	1	28	`PASS (Verified)`
12	`tests/test_browser_broker.py`	591	30,198	6	41	`PASS (Verified)`
13	`tests/test_capabilities_summary.py`	53	3,122	1	4	`PASS (Verified)`
14	`tests/test_categories.py`	56	2,933	1	3	`PASS (Verified)`
15	`tests/test_chapters.py`	29	1,545	1	2	`PASS (Verified)`
16	`tests/test_cli_startup.py`	41	1,473	1	2	`PASS (Verified)`
17	`tests/test_closeout.py`	62	2,893	1	6	`PASS (Verified)`
18	`tests/test_codex_update70_guards.py`	512	27,730	9	25	`PASS (Verified)`
19	`tests/test_collection_bilibili.py`	230	11,547	6	20	`PASS (Verified)`
20	`tests/test_collection_members.py`	143	7,836	2	22	`PASS (Verified)`
21	`tests/test_collections_dom.py`	636	37,306	7	46	`PASS (Verified)`
22	`tests/test_collections.py`	187	10,823	1	31	`PASS (Verified)`

#	Test Module Path (publishops-crosscast/tests/)	Test LOC	Bytes	Classes	Test Funcs	Status
23	tests/test_comment_actions_dom.py	33	2,681	1	4	PASS (Verified)
24	tests/test_comment_snapshot.py	242	14,436	5	26	PASS (Verified)
25	tests/test_cover_prerequisites.py	118	7,327	1	7	PASS (Verified)
26	tests/test_covers.py	199	10,391	1	14	PASS (Verified)
27	tests/test_declarations.py	188	8,634	2	9	PASS (Verified)
28	tests/test_deletion.py	130	7,724	1	4	PASS (Verified)
29	tests/test_diagnostics.py	133	7,688	4	9	PASS (Verified)
30	tests/test_docs.py	172	8,952	1	10	PASS (Verified)
31	tests/test_douyin_cover_read.py	188	8,581	1	11	PASS (Verified)
32	tests/test_douyin_form.py	757	47,314	12	69	PASS (Verified)
33	tests/test_edits.py	196	12,250	4	20	PASS (Verified)
34	tests/test_engagement_dom.py	55	4,386	1	7	PASS (Verified)
35	tests/test_engagement.py	96	6,623	2	13	PASS (Verified)
36	tests/test_field_fixes.py	433	21,895	15	42	PASS (Verified)
37	tests/test_file_selection_guards.py	179	7,259	5	5	PASS (Verified)
38	tests/test_final_review_fixes.py	252	14,099	7	20	PASS (Verified)
39	tests/test_first_run.py	167	9,516	1	15	PASS (Verified)
40	tests/test_free_features.py	894	49,274	13	84	PASS (Verified)
41	tests/test_hidden_desktop_lifecycle.py	372	18,176	5	22	PASS (Verified)
42	tests/test_hostile_messages.py	69	3,167	1	2	PASS (Verified)
43	tests/test_instagram_comment_delete.py	236	13,142	3	21	PASS (Verified)
44	tests/test_instagram_media.py	69	3,718	1	4	PASS (Verified)
45	tests/test_installer.py	151	8,250	1	6	PASS (Verified)
46	tests/test_language_policy.py	33	1,970	1	4	PASS (Verified)
47	tests/test_less_waiting.py	213	7,994	10	14	PASS (Verified)
48	tests/test_malformed_input.py	154	9,522	2	7	PASS (Verified)
49	tests/test_metadata_cover_playlist.py	143	7,783	1	6	PASS (Verified)
50	tests/test_operations_011.py	198	13,036	3	23	PASS (Verified)
51	tests/test_operations_070.py	131	9,033	1	16	PASS (Verified)
52	tests/test_operations_080.py	177	12,290	5	21	PASS (Verified)
53	tests/test_operations_dom.py	51	3,569	1	3	PASS (Verified)
54	tests/test_operations_douyin.py	48	3,211	2	5	PASS (Verified)
55	tests/test_operations_sync.py	90	6,040	1	10	PASS (Verified)
56	tests/test_operations.py	144	9,403	3	16	PASS (Verified)
57	tests/test_overview_lookup_u83.py	89	4,637	2	7	PASS (Verified)
58	tests/test_overview.py	583	34,012	5	58	PASS (Verified)
59	tests/test_paid_api_hook.py	66	3,884	1	5	PASS (Verified)
60	tests/test_paid_features_hook.py	52	2,498	1	5	PASS (Verified)
61	tests/test_platform_warnings.py	370	22,178	7	32	PASS (Verified)
62	tests/test_preferences.py	232	12,274	4	19	PASS (Verified)
63	tests/test_progress.py	659	30,720	15	38	PASS (Verified)
64	tests/test_public_text.py	193	11,772	6	22	PASS (Verified)
65	tests/test_publication_review.py	209	11,309	3	13	PASS (Verified)
66	tests/test_publish_checkpoint_dedup.py	164	7,518	1	6	PASS (Verified)
67	tests/test_publish_plan_safety.py	386	20,346	7	29	PASS (Verified)
68	tests/test_publish_refill.py	334	15,801	1	7	PASS (Verified)
69	tests/test_publish_workers.py	697	34,777	4	25	PASS (Verified)
70	tests/test_rehearsal.py	175	9,632	1	12	PASS (Verified)
71	tests/test_review_consent_independent.py	120	5,918	1	4	PASS (Verified)
72	tests/test_review_followup.py	157	7,963	1	7	PASS (Verified)
73	tests/test_runtime_cleanup.py	283	15,241	1	17	PASS (Verified)
74	tests/test_safety_processes.py	285	13,677	5	18	PASS (Verified)
75	tests/test_safety.py	510	30,138	9	55	PASS (Verified)
76	tests/test_stage1_bilibili_pacing.py	345	16,085	10	19	PASS (Verified)
77	tests/test_stage1_collections.py	263	15,897	3	23	PASS (Verified)
78	tests/test_stage1_edits.py	617	38,835	7	50	PASS (Verified)

#	Test Module Path (publishops-crosscast/tests/)	Test LOC	Bytes	Classes	Test Funcs	Status
79	tests/test_stage1_likes.py	378	20,066	7	36	PASS (Verified)
80	tests/test_stage1_limits.py	173	10,693	3	16	PASS (Verified)
81	tests/test_stage1_messages.py	270	14,486	5	20	PASS (Verified)
82	tests/test_stage1_publish.py	888	50,505	9	94	PASS (Verified)
83	tests/test_stage1_reading.py	365	22,871	12	38	PASS (Verified)
84	tests/test_stage1_repeats.py	351	18,114	3	16	PASS (Verified)
85	tests/test_stage1_replies.py	458	24,230	6	33	PASS (Verified)
86	tests/test_stage1_reporting.py	107	6,256	3	9	PASS (Verified)
87	tests/test_stage1_stops.py	128	6,108	4	8	PASS (Verified)
88	tests/test_stage2_incoming.py	330	17,281	8	25	PASS (Verified)
89	tests/test_stage2_output.py	224	12,412	5	17	PASS (Verified)
90	tests/test_stage2_pages.py	159	10,650	4	17	PASS (Verified)
91	tests/test_stage2_shapes.py	307	18,226	6	26	PASS (Verified)
92	tests/test_two_commands.py	460	25,298	1	37	PASS (Verified)
93	tests/test_work_delete_bili.py	79	3,758	1	5	PASS (Verified)
94	tests/test_work_delete_coexistence.py	254	12,722	1	8	PASS (Verified)
95	tests/test_work_delete_douyin.py	472	21,388	4	36	PASS (Verified)
96	tests/test_work_delete_instagram_layout.py	1,038	54,402	10	41	PASS (Verified)
97	tests/test_work_delete_instagram.py	416	22,303	2	30	PASS (Verified)
98	tests/test_work_delete_paths_review.py	291	14,229	1	12	PASS (Verified)
99	tests/test_work_delete_paths.py	286	14,762	1	14	PASS (Verified)
100	tests/test_work_delete_x_dialogs.py	408	18,734	2	20	PASS (Verified)
101	tests/test_work_delete_x_loading.py	240	11,565	2	13	PASS (Verified)
102	tests/test_work_delete_x_readback.py	437	21,475	4	25	PASS (Verified)
103	tests/test_work_delete_x_timeline_contract.py	127	6,061	1	9	PASS (Verified)
104	tests/test_work_delete_x_timeline.py	420	19,708	8	21	PASS (Verified)
105	tests/test_work_delete_x.py	854	41,611	9	34	PASS (Verified)
106	tests/test_work_delete_youtube_absence.py	72	3,605	1	5	PASS (Verified)
107	tests/test_work_delete_youtube_menu.py	65	2,840	2	4	PASS (Verified)
108	tests/test_work_delete_youtube_query.py	38	2,211	1	2	PASS (Verified)
109	tests/test_work_delete_youtube.py	731	32,306	7	30	PASS (Verified)
110	tests/test_work_delete.py	491	27,887	1	27	PASS (Verified)
111	tests/test_work_discovery_wait.py	391	21,972	4	36	PASS (Verified)
112	tests/test_workflow.py	679	38,450	1	50	PASS (Verified)
113	tests/test_write_answers.py	172	10,266	7	26	PASS (Verified)
114	tests/test_x_composer.py	65	3,344	1	4	PASS (Verified)
115	tests/test_x_discovery.py	386	21,118	2	27	PASS (Verified)
116	tests/test_x_edit_consistency.py	282	19,018	2	19	PASS (Verified)
117	tests/test_x_edit_length.py	382	21,414	1	27	PASS (Verified)
118	tests/test_x_post_length.py	182	10,626	4	23	PASS (Verified)
119	tests/test_x_schedule.py	906	61,937	11	93	PASS (Verified)
120	tests/test_x_site.py	517	35,367	9	34	PASS (Verified)
121	tests/test_youtube_browser.py	1,236	72,978	13	100	PASS (Verified)
122	tests/test_youtube_picker.py	190	12,588	3	15	PASS (Verified)
123	tests/test_youtube_playlist_empty.py	164	9,412	1	10	PASS (Verified)
124	tests/test_youtube_playlists.py	1,034	70,870	8	101	PASS (Verified)
125	tests/test_youtube_schedule_zone.py	229	13,867	6	16	PASS (Verified)
126	tests/test_youtube_studio.py	532	29,397	9	39	PASS (Verified)

#	Source Module Path (publishops-crosscast/src/)	Source LOC	Bytes	Classes	Functions
1	src/bili_cli/__init__.py	2	23	0	0
2	src/bili_cli/asr/__init__.py	9	530	0	0
3	src/bili_cli/asr/audio.py	61	2,473	0	3
4	src/bili_cli/asr/engines.py	277	12,538	6	24
5	src/bili_cli/asr/pipeline.py	49	1,963	0	2
6	src/bili_cli/asr/segment.py	118	4,119	1	7

#	Source Module Path (publishops-crosscast/src/)	Source LOC	Bytes	Classes	Functions
7	src/bili_cli/asr/settings.py	50	1,296	0	5
8	src/bili_cli/asr/setup.py	190	8,038	0	9
9	src/bili_cli/cards.py	70	2,028	0	6
10	src/bili_cli/commands/__init__.py	1	0	0	0
11	src/bili_cli/commands/asr.py	149	7,473	0	8
12	src/bili_cli/commands/auth.py	253	10,248	0	13
13	src/bili_cli/commands/autoreply.py	210	9,777	0	19
14	src/bili_cli/commands/comment.py	532	23,799	0	34
15	src/bili_cli/commands/danmaku.py	411	23,220	0	20
16	src/bili_cli/commands/data.py	475	24,349	0	28
17	src/bili_cli/commands/dm_watch.py	305	16,459	0	15
18	src/bili_cli/commands/dm.py	294	14,325	0	17
19	src/bili_cli/commands/dynamic.py	248	11,465	0	15
20	src/bili_cli/commands/feed.py	347	14,620	0	30
21	src/bili_cli/commands/hot.py	129	5,431	0	10
22	src/bili_cli/commands/mcp.py	183	7,044	0	16
23	src/bili_cli/commands/me.py	244	10,541	0	18
24	src/bili_cli/commands/owner.py	60	1,961	0	7
25	src/bili_cli/commands/search.py	112	3,663	0	5
26	src/bili_cli/commands/season.py	203	9,684	0	16
27	src/bili_cli/commands/skills.py	118	5,025	0	11
28	src/bili_cli/commands/upload.py	955	50,258	0	45
29	src/bili_cli/commands/user.py	229	9,291	0	15
30	src/bili_cli/commands/video.py	663	30,998	0	42
31	src/bili_cli/config.py	139	4,072	0	18
32	src/bili_cli/errors.py	133	6,051	1	4
33	src/bili_cli/frames.py	318	14,598	0	13
34	src/bili_cli/guard.py	129	4,837	0	6
35	src/bili_cli/health.py	156	5,877	1	11
36	src/bili_cli/im_push.py	172	6,232	2	12
37	src/bili_cli/main.py	166	6,134	0	8
38	src/bili_cli/mcp_server.py	188	10,840	0	22
39	src/bili_cli/output.py	60	1,673	1	6
40	src/bili_cli/reactions.py	129	5,358	0	7
41	src/bili_cli/runtime.py	244	8,749	0	13
42	src/bili_cli/skills.py	228	8,795	1	15
43	src/video_publisher/__init__.py	4	103	0	0
44	src/video_publisher/__main__.py	5	35	0	0
45	src/video_publisher/adapters.py	359	18,092	0	26
46	src/video_publisher/auth_jobs.py	234	10,664	0	14
47	src/video_publisher/browser_backends.py	995	63,951	1	70
48	src/video_publisher/browser_broker.py	387	17,945	0	23
49	src/video_publisher/browser_edits.py	476	30,811	0	31
50	src/video_publisher/browser_forms.py	1,234	83,395	0	70
51	src/video_publisher/browser_runtime.py	529	31,890	0	36
52	src/video_publisher/browser_worker.py	114	6,170	0	4
53	src/video_publisher/categories.py	15	993	0	1
54	src/video_publisher/chapters.py	70	3,991	0	6
55	src/video_publisher/cli.py	1,434	113,979	0	37
56	src/video_publisher/collections_bilibili.py	96	4,465	0	8
57	src/video_publisher/collections_browser.py	448	28,244	0	20
58	src/video_publisher/collections_douyin.py	184	13,062	0	7
59	src/video_publisher/collections_instagram.py	88	5,817	0	3
60	src/video_publisher/collections.py	225	14,188	0	13
61	src/video_publisher/comment_actions_api.py	62	4,240	0	1
62	src/video_publisher/comment_actions_browser.py	125	8,020	0	5

#	Source Module Path (publishops-crosscast/src/)	Source LOC	Bytes	Classes	Functions
63	src/video_publisher/comment_actions_douyin.py	87	5,380	0	3
64	src/video_publisher/comment_actions_x.py	91	6,529	0	2
65	src/video_publisher/comment_actions.py	236	16,801	0	11
66	src/video_publisher/comment_collection.py	82	4,088	0	4
67	src/video_publisher/core.py	655	40,870	2	44
68	src/video_publisher/covers.py	221	13,974	0	5
69	src/video_publisher/deletion.py	117	7,384	0	5
70	src/video_publisher/direct_messages_bili.py	37	2,408	0	1
71	src/video_publisher/direct_messages_browser.py	114	7,991	0	4
72	src/video_publisher/direct_messages_douyin.py	58	3,675	0	1
73	src/video_publisher/direct_messages_x.py	120	7,444	0	6
74	src/video_publisher/direct_messages.py	156	9,382	0	12
75	src/video_publisher/drafts_bilibili.py	95	5,642	0	6
76	src/video_publisher/edits_bilibili.py	184	11,291	0	12
77	src/video_publisher/edits.py	238	18,163	0	10
78	src/video_publisher/engagement_api.py	65	3,483	0	5
79	src/video_publisher/engagement_browser.py	116	6,667	0	6
80	src/video_publisher/engagement.py	147	9,250	0	9
81	src/video_publisher/hidden_desktop.py	175	10,176	4	9
82	src/video_publisher/instagram_media.py	60	3,346	0	1
83	src/video_publisher/language_policy.py	20	1,051	0	1
84	src/video_publisher/mentions_browser.py	126	7,171	0	4
85	src/video_publisher/message_sources.py	62	3,721	0	2
86	src/video_publisher/messages.py	103	6,678	0	8
87	src/video_publisher/metadata.py	133	11,966	0	2
88	src/video_publisher/native_browser.py	397	29,666	0	9
89	src/video_publisher/operations_api.py	150	9,989	0	7
90	src/video_publisher/operations_batch.py	70	4,198	0	3
91	src/video_publisher/operations_browser.py	355	24,390	0	9
92	src/video_publisher/operations_douyin.py	101	6,194	0	4
93	src/video_publisher/operations_inbox.py	60	2,802	0	5
94	src/video_publisher/operations_runs.py	376	22,152	0	20
95	src/video_publisher/operations_sync.py	113	7,322	0	4
96	src/video_publisher/operations.py	294	22,195	0	14
97	src/video_publisher/overview.py	402	22,077	0	27
98	src/video_publisher/preferences.py	54	2,597	0	4
99	src/video_publisher/process_lock.py	99	4,028	0	3
100	src/video_publisher/progress.py	134	5,254	0	8
101	src/video_publisher/publication_review.py	110	5,112	0	8
102	src/video_publisher/quick.py	195	14,031	0	7
103	src/video_publisher/read_scheduler.py	72	3,239	0	3
104	src/video_publisher/rehearsal.py	143	6,910	0	8
105	src/video_publisher/runtime_delete.py	75	4,030	1	2
106	src/video_publisher/runtime_files.py	209	9,532	0	8
107	src/video_publisher/runtime_scratch.py	80	3,579	0	3
108	src/video_publisher/safety.py	381	23,971	0	31
109	src/video_publisher/widgets_bilibili.py	386	21,290	0	25
110	src/video_publisher/work_catalog.py	79	4,661	0	3
111	src/video_publisher/work_delete_bili.py	24	1,695	0	1
112	src/video_publisher/work_delete_browser.py	18	1,062	0	1
113	src/video_publisher/work_delete_douyin.py	263	13,508	0	13
114	src/video_publisher/work_delete_instagram.py	439	25,741	1	22
115	src/video_publisher/work_delete_state.py	183	8,989	0	12
116	src/video_publisher/work_delete_x_readback.py	207	9,597	1	11
117	src/video_publisher/work_delete_x.py	320	16,828	0	15
118	src/video_publisher/work_delete_youtube.py	307	15,877	1	20

#	Source Module Path (`publishops-crosscast/src/`)	Source LOC	Bytes	Classes	Functions
119	`src/video_publisher/work_delete.py`	154	9,329	0	5
120	`src/video_publisher/work_discovery_browser.py`	114	6,939	0	3
121	`src/video_publisher/x_pages.py`	105	6,807	0	7
122	`src/video_publisher/youtube_playlists.py`	38	2,024	0	4
123	`src/video_publisher/youtube_studio.py`	997	59,535	0	54

Appendix F. Primary Machine-Readable Audit Receipts & Production Update Logs

Below are the verbatim machine-readable JSON audit receipts from `data/scheduled-run-evidence.json`, the 24-hour post-publication checkpoints, and the longitudinal daily production updates (`2026-10-28` through `2026-11-03`).

File: `data/scheduled-run-evidence.json`


```
{
  "checked_at": "2026-10-28T04:40:24.721Z",
  "status": "VERIFIED_AUTOMATIC_RUN_COMPLETED",
  "trigger": "automation_heartbeat_scheduled",
  "planned_at": "2026-10-28T12:37:13.943+08:00",
  "started_at": "2026-10-28T04:37:38.935Z",
  "completed_at": "2026-10-28T04:38:10.110Z",
  "duration_ms": 31175,
  "trigger_delay_ms": 24992,
  "work_completed": [
    "Read instance ownership, operational checklist, stop conditions and current checkpoints.",
    "Verified comments, interactive components, hot research and analytics were not due.",
    "Synchronized latest user instructions into the operational checklist, review playbook and automation prompt.",
    "No Bilibili platform operations were needed in this run."
  ],
  "task_errors": [],
  "manual_test_triggered": false,
  "duplicate_run_started": false
}
```


File: `data/episode03-24h-2026-10-28.json`


```
{
  "source": "Recorded creator analytics checkpoint",
  "bvid": "BV1UXxN6aECf",
  "published_at": "2026-10-27T19:30:00+08:00",
  "due_at": "2026-10-28T19:30:00+08:00",
  "sampled_at": "2026-10-28T19:53:00+08:00",
  "age_bucket": "24h",
  "actual_age_hours": 24.383333333333333,
  "duration_seconds": 70,
  "metrics": {
    "view": 229,
    "like": 8,
    "favorite": 2,
    "coin": 0,
    "reply": 7,
    "share": 0,
    "danmaku": 0,
    "completion_percent": 44.64,
    "avg_watch_seconds": 31,
    "per_100_plays": {
      "like": 3.49,
      "favorite": 0.87,
      "coin": 0,
      "reply": 3.06,
      "share": 0,
      "danmaku": 0
    }
  },
  "missing": [],
  "limitation_zh_hant": "到期後 23 分鐘採樣；單片描述性數據，無法證明模型或題材造成增長。",
  "limitation_en": "Sampled 23 minutes after the due time; descriptive single-video data cannot establish causal model or topic effects."
}
```


File: `data/production-update-2026-10-28-evening.json`


```
{
  "date": "2026-10-28",
  "videos": [
    {
      "bvid": "BV1C4xE6oEjS",
      "title_zh_hant": "鯨魚娘代班日誌 04：幫個小忙",
      "title_en": "Whale Girl Shift Log 04: small favours",
      "state": "scheduled_on_bilibili",
      "scheduled_time": "2026-11-01T12:00:00+08:00",
      "source_video_sha256": "c23c171149bc47900a3b574b15fb01c873059ccc5e9a166c834d870bcbc86bb3",
      "duration_seconds": 132
    },
    {
      "bvid": "BV1Pkx76ZEsh",
      "title_zh_hant": "鯨魚娘代班日誌 05：帳單",
      "title_en": "Whale Girl Shift Log 05: the bill",
      "state": "scheduled_on_bilibili",
      "scheduled_time": "2026-11-01T19:30:00+08:00",
      "source_video_sha256": "f735aca1bb770a16cbcfbabc7ecaaac37e19b14498f9279d717047e57e223901",
      "duration_seconds": 148.2
    },
```

```
    {
      "bvid": "BV1Mvx76YEty",
      "title_zh_hant": "GPT 值班日記 05：Seedance 入門（閃圖修正版）",
      "title_en": "GPT Duty Log 05: Seedance tutorial (flash correction)",
      "state": "scheduled_on_bilibili",
      "scheduled_time": "2026-10-30T19:30:00+08:00",
      "source_video_sha256": "5dc121e833dcfc5a52ea0c3077b6b4e19e80815b45ba53ba6a3133ed3f5bda6b",
      "duration_seconds": 434.466666667
    }
  ],
  "whale02": {
    "bvid": "BV1Wfx56JEih",
    "published_at": "2026-10-28T19:30:00+08:00",
    "confirmed_by": "Post-publication receipt and current creator-list readback"
  },
  "excluded_work_in_progress": {
    "label": "GPT episode 06",
    "reason": "RENDER_OR_QA_PENDING at the daily check; no final delivery published here"
  }
}
```


File: `data/production-update-2026-10-29-evening.json`


```
{
  "sampled_at": "2026-10-29T19:39:00+08:00",
  "source": "Local delivery and operations receipts; no new platform writes or queries during this update",
  "completed_files": [
    {
      "file": "gpt-log-06-v2-20261029.mp4",
      "sha256": "b53d3df6f85b2e361af10636eacfcb88a5cd672401028fd133dfa403e07b5454",
      "bytes": 44282436,
      "duration_seconds": 591.466667,
      "bvid": "BV1iBs96VEL9",
      "scheduled_time": "2026-10-31T12:00:00+08:00",
      "title_original": "GPT值班日记06丨手把手教你如何让DeepSeek帮我打工",
      "state": "APPROVED_SCHEDULED_AWAITING_PUBLIC"
    },
    {
      "file": "whale-log-06-v2-20261029.mp4",
      "sha256": "35bc463bc4c7fca56aea2fcf9addd55b8ca19dc74231f6aa03dcaa0dfd0d35ba",
      "bytes": 14552237,
      "duration_seconds": 184.2,
      "bvid": "BV1jis96yENU",
      "scheduled_time": "2026-11-02T12:00:00+08:00",
      "title_original": "鲸鱼娘代班日志06丨AI负责全自动，我负责假装人工",
      "state": "APPROVED_SCHEDULED_WITH_RESERVATION_AWAITING_PUBLIC"
    }
  ],
  "whale_log03": {
    "bvid": "BV1Wcx76dErs",
    "state": "published",
    "actual_public_time": "2026-10-29T19:30:00+08:00",
    "checked_at": "2026-10-29T19:30:36+08:00"
  },
  "crosscast": {
    "version": "0.3.0-alpha",
    "verified_commit": "596e9517dc02e19637bd8115fd1108279dbb0540",
    "offline_tests_previously_passed": 86,
    "tests_rerun_this_update": false,
    "episode02_submission_verified": [
      "youtube",
      "instagram",
      "x"
    ],
    "douyin": "Submitted and review card observed; auth expired after restart; final readback pending"
  },
  "privacy_review": {
    "gpt06": "Production review records 13 source windows and 26 boundary frames within approved sanitized ranges; v2 receipt records safe media
unchanged. Raw recordings excluded.",
    "whale06": "Rendered character animation; local delivery and technical QA records only."
  },
  "acceptance_limit": "Platform review approval and technical QA are separate from full-film human acceptance; no continuous human viewing or lis
tening claim.",
  "historical_snapshots_preserved": true
}
```


File: `data/production-update-2026-10-30-evening.json`


```
{
  "sampled_at": "2026-10-30T12:58:08.002Z",
  "source": "Local production delivery and dated operations receipts; no new platform queries or actions.",
  "completed_local_deliveries": [
    {
      "file": "study-whale-transformation-v3-20261030.mp4",
      "title_zh_hant": "鯨魚娘變身實驗 v3（較早交付）",
      "title_en": "Whale Girl transformation study v3 (earlier delivery)",
      "version": "v3",
      "bytes": 90012465,
      "sha256": "60f03a12439c0c05d9125ea455ed66f194b51695155a1c1bfce9f9ca07486c25",
      "duration_seconds": 67.0,
      "frames": 2010,
      "bilibili_state_at_snapshot": "unpublished_experiment",
      "bvid": null,
      "bilibili_current_state": "unpublished_experiment",
      "github_asset_status": "published_verified",
      "state_verified_at": "2026-10-30T20:39:25+08:00",
      "acceptance": "technical_delivery_not_full_user_acceptance",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-10-30-evening/study-whale-transformation-v
3-20261030.mp4"
    },
    {
```

```
      "file": "study-whale-transformation-v4-20261030.mp4",
      "title_zh_hant": "鯨魚娘變身實驗 v4（定向修正）",
      "title_en": "Whale Girl transformation study v4 (targeted repair)",
      "version": "v4",
      "bytes": 103106107,
      "sha256": "3b209af916b128f63ff1e0e9daae5a7a613b0def7ace11610bb318556448a99c",
      "duration_seconds": 75.23333333333333,
      "frames": 2257,
      "bilibili_state_at_snapshot": "unpublished_experiment",
      "bvid": null,
      "bilibili_current_state": "unpublished_experiment",
      "github_asset_status": "published_verified",
      "state_verified_at": "2026-10-30T20:39:25+08:00",
      "acceptance": "technical_delivery_not_full_user_acceptance",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-10-30-evening/study-whale-transformation-v
4-20261030.mp4"
    }
  ],
  "github_media_upload": "PUBLISHED_AND_VERIFIED",
  "gpt_duty04": {
    "bvid": "BV1xFx764EjN",
    "state": "published",
    "actual_public_time": "2026-10-30T12:00:00+08:00",
    "checked_at": "2026-10-30T13:00:00+08:00",
    "chapters_count": 8,
    "chapters_state": "SET_READBACK_CONFIRMED_PENDING_AUDIT",
    "personalized_components": "FOLLOW_VOTE_LINK_RATING_READBACK_CONFIRMED",
    "collection_readback": "STALE_OR_DELAYED"
  },
  "v4_technical_evidence": {
    "resolution": "1080x1920",
    "fps": 30,
    "frames": 2257,
    "full_decode": "passed",
    "clipped_samples": 0,
    "peak_sample_dbfs": -1.6361967778880953,
    "silence_window_seconds": [
      50.763333333333335,
      62.50333333333334
    ],
    "full_black_at_seconds": 61.5,
    "picture_stream_bit_identical_after_audio_mix": true,
    "human_continuous_watch": false,
    "human_full_listening": false
  },
  "crosscast": {
    "version": "0.15.1",
    "verified_commit": "91615aa71e250ae0ec027d00c310d3e016ad35d4",
    "historical_offline_tests": 417,
    "tests_rerun_this_update": false
  },
  "analytics_scope": "No newer full-channel/24-hour sample was found; previous dated observations retained, no zero-fill or extrapolation.",
  "excluded": [
    "raw screen recordings",
    "third-party reference originals",
    "private account logs and messages",
    "intermediate segment renders",
    "unrelated personal materials"
  ],
  "historical_snapshots_preserved": true,
  "github_published_verified_at": "2026-11-01T13:11:05.166937+00:00"
}
```


File: `data/production-update-2026-11-01.json`


```
{
  "checked_at": "2026-11-01T13:11:05.166937+00:00",
  "source_check_started_at": "2026-11-01T13:02:52.315Z",
  "source": "Completed local delivery receipts and sanitized CrossCast acceptance records; GitHub read-back; zero Bilibili/YouTube/Douyin/Instagr
am/X operations in this maintenance run.",
  "new_media": [
    {
      "file": "study-whale-transformation-v3-20261030.mp4",
      "title_zh_hant": "鯨魚娘變身實驗 v3（較早交付）",
      "title_en": "Whale Girl transformation study v3 (earlier delivery)",
      "version": "v3",
      "bytes": 90012465,
      "sha256": "60f03a12439c0c05d9125ea455ed66f194b51695155a1c1bfce9f9ca07486c25",
      "duration_seconds": 67.0,
      "frames": 2010,
      "bilibili_state_at_snapshot": "unpublished_experiment",
      "bvid": null,
      "bilibili_current_state": "unpublished_experiment",
      "github_asset_status": "published_verified",
      "state_verified_at": "2026-10-30T20:39:25+08:00",
      "acceptance": "technical_delivery_not_full_user_acceptance",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-10-30-evening/study-whale-transformation-v
3-20261030.mp4",
      "github_verified_at": "2026-11-01T13:11:05.166937+00:00"
    },
    {
      "file": "study-whale-transformation-v4-20261030.mp4",
      "title_zh_hant": "鯨魚娘變身實驗 v4（定向修正）",
      "title_en": "Whale Girl transformation study v4 (targeted repair)",
      "version": "v4",
      "bytes": 103106107,
      "sha256": "3b209af916b128f63ff1e0e9daae5a7a613b0def7ace11610bb318556448a99c",
      "duration_seconds": 75.23333333333333,
      "frames": 2257,
      "bilibili_state_at_snapshot": "unpublished_experiment",
      "bvid": null,
      "bilibili_current_state": "unpublished_experiment",
      "github_asset_status": "published_verified",
```

```
      "state_verified_at": "2026-10-30T20:39:25+08:00",
      "acceptance": "technical_delivery_not_full_user_acceptance",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-10-30-evening/study-whale-transformation-v
4-20261030.mp4",
      "github_verified_at": "2026-11-01T13:11:05.166937+00:00"
    },
    {
      "file": "study-whale-color-heels-r5-20261031.mp4",
      "title_zh_hant": "鯨魚娘變身實驗：局部調色與腳步聲修訂5",
      "title_en": "Whale Girl study: local colour and heel-sound revision 5",
      "bytes": 113352299,
      "sha256": "19ea894908ce3600d2fecd2f7ed796154ba17988b7fc22b9d0bc617efa383e50",
      "duration_seconds": 66.866667,
      "frames": 2006,
      "bvid": null,
      "bilibili_state_at_snapshot": "not_verified",
      "bilibili_current_state": "not_verified",
      "github_asset_status": "published_verified",
      "acceptance": "completed_experimental_delivery_pending_full_user_review",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-01-studies/study-whale-color-heels-r5-2
0261031.mp4",
      "github_verified_at": "2026-11-01T13:11:05.166937+00:00"
    },
    {
      "file": "study-whale-motion-r7-20261101.mp4",
      "title_zh_hant": "鯨魚娘變身實驗：動作節奏修訂7",
      "title_en": "Whale Girl study: motion-timing revision 7",
      "bytes": 108503225,
      "sha256": "933e1643a7351912fdb38f1e2f8894b8e90031d7852545226c0c9a13962c327e",
      "duration_seconds": 65.366667,
      "frames": 1961,
      "bvid": null,
      "bilibili_state_at_snapshot": "not_verified",
      "bilibili_current_state": "not_verified",
      "github_asset_status": "published_verified",
      "acceptance": "completed_experimental_delivery_pending_full_user_review",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-01-studies/study-whale-motion-r7-202611
01.mp4",
      "github_verified_at": "2026-11-01T13:11:05.166937+00:00"
    }
  ],
  "verified_public_video_versions": 25,
  "new_analytics_sample": false,
  "historical_metrics_unchanged": true,
  "crosscast": {
    "portfolio_code_snapshot": "0.15.1",
    "pull_request": 3,
    "pr_state_at_check": "OPEN",
    "public_branch_head_at_check": "80e166b62f5061904e5c8b63699e2940a6866bb8",
    "local_candidate_only": "988bee228dcfaf60ee3378ec36a31fbd59916cc7",
    "candidate_code_imported": false,
    "merged": false,
    "ci": {
      "commit": "eaa1aab128c6eb5aa9cc09036872f8e111da1969",
      "run_url": "https://github.com/whitecarp/publishops-crosscast/actions/runs/41301333026",
      "python": [
        "3.10",
        "3.11",
        "3.12"
      ],
      "result": "success"
    },
    "offline_test_runs": [
      {
        "commit": "80e166b",
        "tests": 143,
        "result": "passed"
      },
      {
        "commit": "988bee2",
        "tests": 13,
        "result": "passed",
        "independent_pin_counterexamples": "failed"
      }
    ],
    "acceptance": "partial_existing_feature_closeout_not_release_ready"
  },
  "pending_source_groups": [
    {
      "group": "episode-seven artwork and cast-v7",
      "reason": "production assets without final film delivery; excluded this run"
    }
  ],
  "selection_note": "Newest complete delivery from each active revision branch, plus two previously prepared archive exports. Superseded intermed
iate revisions and isolated segment renders remain local.",
  "excluded": [
    "credentials",
    "private messages",
    "viewer personal data",
    "raw client logs",
    "raw screen recordings",
    "standalone third-party reference originals",
    "in-progress assets"
  ]
}
```


File: `data/production-update-2026-11-02.json`


```
{
  "snapshot_at": "2026-11-02T03:37:13.846Z",
  "source_window_start": "2026-11-01T13:02:52.315Z",
  "source": "Completed production files, production QA notes and existing local upload receipts; no platform refresh",
  "new_media": [
```

```
    {
      "file": "whale-log-07-v3-20261101.mp4",
      "title_zh_hant": "鯨魚娘代班日誌07：值班模擬器（v3）",
      "title_en": "Whale Girl Shift Log 07: duty simulator (v3)",
      "bytes": 49307630,
      "sha256": "e37285ee4f8c9fc4fac70978e06638e40203f8337bc286344ddb5731a8814ca6",
      "duration_seconds": 179.76666666666668,
      "frames": 5393,
      "width": 1920,
      "height": 1080,
      "fps": 30,
      "bvid": "BV1aPxt6eEDR",
      "bilibili_state_at_snapshot": "scheduled_submission_recorded",
      "bilibili_current_state": "not_refreshed",
      "scheduled_time": "2026-11-03T12:00:00+08:00",
      "state_verified_at": "2026-11-01T17:01:29.519773+00:00",
      "state_source": "local_publishing_receipt_not_live_refresh",
      "bilibili_part": null,
      "github_asset_status": "published_verified",
      "acceptance": "landscape_delivery_user_overall_review_recorded",
      "validation": {
        "sha256_matches_delivery_receipt": true,
        "full_decode_errors": 0,
        "h264_aac": true
      },
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-02-episode07/whale-log-07-v3-20261101.m
p4",
      "github_verified_at": "2026-11-02T03:49:24.540068+00:00"
    },
    {
      "file": "whale-log-07-cut-a-v3-20261101.mp4",
      "title_zh_hant": "第七期直式切片A：斷糧代班人",
      "title_en": "Episode 07 portrait cut A: running out of resources",
      "bytes": 7861825,
      "sha256": "8429157c4fe4446d24d7b74a2ab354d1e3552b79329a5a52bdcbb12cad09a80e",
      "duration_seconds": 28.1,
      "frames": 843,
      "width": 1080,
      "height": 1920,
      "fps": 30,
      "bvid": "BV1gexh6rEZM",
      "bilibili_state_at_snapshot": "scheduled_submission_recorded",
      "bilibili_current_state": "not_refreshed",
      "scheduled_time": "2026-11-03T19:30:00+08:00",
      "state_verified_at": "2026-11-02T02:53:32.197090+00:00",
      "state_source": "local_publishing_receipt_not_live_refresh",
      "bilibili_part": 3,
      "github_asset_status": "published_verified",
      "acceptance": "complete_portrait_cut_pending_full_motion_and_audio_review",
      "validation": {
        "sha256_matches_delivery_receipt": true,
        "full_decode_errors": 0,
        "h264_aac": true
      },
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-02-episode07/whale-log-07-cut-a-v3-2026
1101.mp4",
      "github_verified_at": "2026-11-02T03:49:24.540068+00:00"
    },
    {
      "file": "whale-log-07-cut-b-v3-20261101.mp4",
      "title_zh_hant": "第七期直式切片B：差評製造機",
      "title_en": "Episode 07 portrait cut B: refusing every request",
      "bytes": 8185740,
      "sha256": "1e436aff188953b42e9d648d5c2e0a73fb32b849a6b9983b6b1a11be939c33bd",
      "duration_seconds": 29.9,
      "frames": 897,
      "width": 1080,
      "height": 1920,
      "fps": 30,
      "bvid": "BV1gexh6rEZM",
      "bilibili_state_at_snapshot": "scheduled_submission_recorded",
      "bilibili_current_state": "not_refreshed",
      "scheduled_time": "2026-11-03T19:30:00+08:00",
      "state_verified_at": "2026-11-02T02:53:32.197090+00:00",
      "state_source": "local_publishing_receipt_not_live_refresh",
      "bilibili_part": 1,
      "github_asset_status": "published_verified",
      "acceptance": "complete_portrait_cut_pending_full_motion_and_audio_review",
      "validation": {
        "sha256_matches_delivery_receipt": true,
        "full_decode_errors": 0,
        "h264_aac": true
      },
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-02-episode07/whale-log-07-cut-b-v3-2026
1101.mp4",
      "github_verified_at": "2026-11-02T03:49:24.540068+00:00"
    },
    {
      "file": "whale-log-07-cut-c-v3-20261101.mp4",
      "title_zh_hant": "第七期直式切片C：幻覺藝術家",
      "title_en": "Episode 07 portrait cut C: bluffing through the shift",
      "bytes": 7536390,
      "sha256": "e7ba91f7e903e27e35ac40b57d2ea8d8c40a910ce5025fb01a35e1151f8892a3",
      "duration_seconds": 29.966666666666665,
      "frames": 899,
      "width": 1080,
      "height": 1920,
      "fps": 30,
      "bvid": "BV1gexh6rEZM",
      "bilibili_state_at_snapshot": "scheduled_submission_recorded",
      "bilibili_current_state": "not_refreshed",
      "scheduled_time": "2026-11-03T19:30:00+08:00",
      "state_verified_at": "2026-11-02T02:53:32.197090+00:00",
      "state_source": "local_publishing_receipt_not_live_refresh",
```

```
      "bilibili_part": 2,
      "github_asset_status": "published_verified",
      "acceptance": "complete_portrait_cut_pending_full_motion_and_audio_review",
      "validation": {
        "sha256_matches_delivery_receipt": true,
        "full_decode_errors": 0,
        "h264_aac": true
      },
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-02-episode07/whale-log-07-cut-c-v3-2026
1101.mp4",
      "github_verified_at": "2026-11-02T03:49:24.540068+00:00"
    }
  ],
  "verified_public_video_versions": 29,
  "new_analytics_sample": false,
  "historical_metrics_unchanged": true,
  "bilibili_receipt_note": "Submission and scheduled time are historical receipt fields, not current-publication verification. B/C/A multipart fi
les are deduplicated by SHA-256.",
  "historical_standalone_b_submission": {
    "bvid": "BV1F5xh6sE82",
    "recorded_at": "2026-11-02T02:23:01.059508+00:00",
    "current_state": "not_refreshed"
  },
  "crosscast": {
    "portfolio_code_snapshot": "0.15.1",
    "tested_commit": "2d54c8fe485ed58a81b7bdd049cd8d258b7e453f",
    "ci": {
      "commit": "2d54c8fe485ed58a81b7bdd049cd8d258b7e453f",
      "url": "https://github.com/whitecarp/publishops-crosscast/actions/runs/41380631557",
      "conclusion": "success",
      "jobs": [
        {
          "name": "Python 3.12",
          "conclusion": "success"
        },
        {
          "name": "Python 3.11",
          "conclusion": "success"
        },
        {
          "name": "Python 3.10",
          "conclusion": "success"
        }
      ]
    },
    "pr_state_at_check": "open",
    "pr_head_at_check": "7114451abb51e3446afdffa14feb532a63ab56db",
    "merged": false,
    "candidate_code_imported": false,
    "targeted_offline_checks": 16,
    "x_fresh_upload_preview": "passed_and_media_removed_without_public_post",
    "x_scheduled_submission": "independent_row_observed_and_cancelled_with_media_cleanup",
    "x_cli_receipt": "SUBMITTED_UNVERIFIED",
    "x_durable_media_pairing": "not_proven",
    "x_future_publication": "not_tested",
    "douyin_private_caption": "exact_card_and_editor_no_stray_stars",
    "douyin_edit": "blocked_by_platform_review_before_typing_or_saving",
    "douyin_test_cleanup": "pending_at_snapshot",
    "acceptance": "scoped_results_not_full_release_acceptance"
  },
  "production_limits": [
    "portrait_full_playback_listening_and_device_overlay_checks_pending",
    "landscape_stamp_timing_measurements_disagree",
    "fictional_game_cases_selected_not_average_results"
  ],
  "pending_source_groups": [
    {
      "group": "portrait cuts",
      "reason": "complete exports archived, continuous playback/listening and phone overlay acceptance still pending"
    }
  ],
  "excluded": [
    "raw_screen_recordings",
    "unfinished_assets",
    "unrelated_personal_or_client_work",
    "credentials",
    "private_messages",
    "viewer_personal_data",
    "raw_client_logs"
  ],
  "github_assets_verified_at": "2026-11-02T03:49:24.540068+00:00"
}
```


File: `data/production-update-2026-11-03.json`


```
{
  "snapshot_at": "2026-11-03T04:16:22.732Z",
  "source_window_start": "2026-11-02T03:37:13.846Z",
  "source": "Completed film, production review, existing publication records, user confirmations and GitHub release/PR metadata; no platform refr
esh",
  "new_media": [
    {
      "file": "i-watched-it-for-you-v3-20261102.mp4",
      "title_zh_hant": "我替你看過了（v3）",
      "title_en": "I Watched It For You (v3)",
      "bytes": 285855005,
      "sha256": "7d6fd76090f7a93b2711e15db4bb924f741d3a904a01fc4b7628fc2130dbafdc",
      "duration_seconds": 176.4,
      "width": 1920,
      "height": 1080,
      "fps": 30,
      "bvid": "BV15B4F6hEYr",
      "bilibili_state_at_snapshot": "public_reported_by_production_session",
```

```
      "bilibili_current_state": "not_refreshed",
      "state_source": "existing_production_handoff_not_new_platform_query",
      "state_verified_at": "2026-11-02T21:01:21+08:00",
      "github_asset_status": "published_verified",
      "github_asset_url": "https://github.com/whitecarp/agentic-work-lab/releases/download/update-2026-11-03-film/i-watched-it-for-you-v3-2026110
2.mp4",
      "acceptance": "published_fictional_short; subjective_audio_and_mobile_subtitle_review_not_fully_established",
      "validation": {
        "metadata_source": "production_review",
        "production_full_decode_errors": 0,
        "production_sampled_frames": 37,
        "source_and_staged_sha256_match": true,
        "historical_local_read_discrepancy_resolved": false
      },
      "github_verified_at": "2026-11-03T04:25:41.625026+00:00"
    }
  ],
  "film_platforms": {
    "bilibili": {
      "url": "https://www.bilibili.com/video/BV15B4F6hEYr",
      "evidence": "production_session_report",
      "fresh_check": false
    },
    "youtube": {
      "url": "https://www.youtube.com/watch?v=AzW6vr8nvDM",
      "state": "PUBLIC_VERIFIED",
      "native_state": "SUBMISSION_VERIFIED"
    },
    "x": {
      "url": "https://x.com/White_Carp2000/status/2117261340633161838",
      "state": "PUBLIC_VERIFIED",
      "native_state": "SUBMISSION_VERIFIED",
      "stored_resolution_verified": false
    },
    "douyin": {
      "state": "USER_CONFIRMED_VISIBLE_NATIVE_VERIFY_BLOCKED",
      "native_state": "UNKNOWN",
      "url": null,
      "cover": "user_phone_confirmation"
    },
    "instagram": {
      "url": "https://www.instagram.com/reel/DfQ9iCiCWYN/",
      "native_state": "SUBMISSION_VERIFIED",
      "one_upload_one_share": true,
      "metadata_readback": true,
      "source_aspect_ratio_preserved": true,
      "ai_label": true,
      "sharing_elsewhere": false,
      "cover": "user_confirmed_mobile_replacement",
      "remote_cover_pixels_verified": false
    }
  },
  "instagram_timing_seconds": {
    "source_plan_checks_outside_publish": 0.438,
    "rehearsal_outside_publish": 0.444,
    "auth_preflight_outside_publish": 10.137,
    "publish_total": 412.217,
    "browser_start_to_file_entry": 11.058,
    "after_submit_wait": 380.551,
    "independent_readback": 5.224
  },
  "timing_limits": [
    "single_observation_not_benchmark",
    "transfer_processing_confirmation_not_separated",
    "no_live_parallel_speed_measurement"
  ],
  "crosscast": {
    "portfolio_code_snapshot": "0.15.1",
    "released_version": "0.16.0",
    "release_commit": "a46bd19372ea3c0c7c4672642d4d56db1b1f42c7",
    "release_url": "https://github.com/whitecarp/publishops-crosscast/releases/tag/v0.16.0",
    "main_at_snapshot": "69aa5907894ced4809e99e744d45b12b5ffbda42",
    "merged_prs": [
      8,
      9
    ],
    "pending_prs": [
      10
    ],
    "pending_local_work": [
      "bounded_parallel_publication"
    ],
    "existing_release_assets_changed": false
  },
  "new_analytics_sample": false,
  "historical_metrics_unchanged": true,
  "verified_public_video_versions": 30,
  "excluded": [
    "credentials",
    "raw_client_logs",
    "private_messages",
    "viewer_personal_data",
    "third_party_originals",
    "unfinished_deliveries",
    "embargoed_Claude_chan_videos_before_2026_11_05_noon_Beijing"
  ],
  "github_assets_verified_at": "2026-11-03T04:25:41.625026+00:00"
}
```


File: `data/whale-log02-24h-2026-10-29.json`


```
{
  "bvid": "BV1Wfx56JEih",
```

```
  "title": "鯨鱼娘代班日志02丨上班第一天，我被人类派了三个活",
  "published_at": "2026-10-28T19:30:00+08:00",
  "due_at": "2026-10-29T19:30:00+08:00",
  "sampled_at": "2026-10-29T19:33:00+08:00",
  "age_bucket": "24h",
  "duration_seconds": 117,
  "metrics": {
    "view": 308,
    "like": 24,
    "favorite": 5,
    "coin": 0,
    "reply": 51,
    "share": 0,
    "danmaku": 1,
    "completion_percent": 36.63,
    "avg_watch_seconds": 42,
    "per_100_plays": {
      "like": 7.79,
      "favorite": 1.62,
      "coin": 0.0,
      "reply": 16.56,
      "share": 0.0,
      "danmaku": 0.32
    }
  },
  "missing": [],
  "source": "Recorded creator analytics checkpoint, reused from the operations instance",
  "actual_age_hours": 24.05,
  "limitation_zh_hant": "到期後 3 分鐘採樣；單片描述性數據。回覆數不等於獨立觀眾數，也不能證明模型或題材造成增長。",
  "limitation_en": "Sampled three minutes after the due time. Descriptive single-video data; replies are not unique viewers and do not establish causal model or topic effects."
}
```